Lesson ***Ten***
Continuous Process Improvement

Effective Project Management:
Traditional, Agile, Extreme
7th Edition

Ch16: Establishing and Managing a Continuous Process Improvement Program

Presented by
Bowen DU

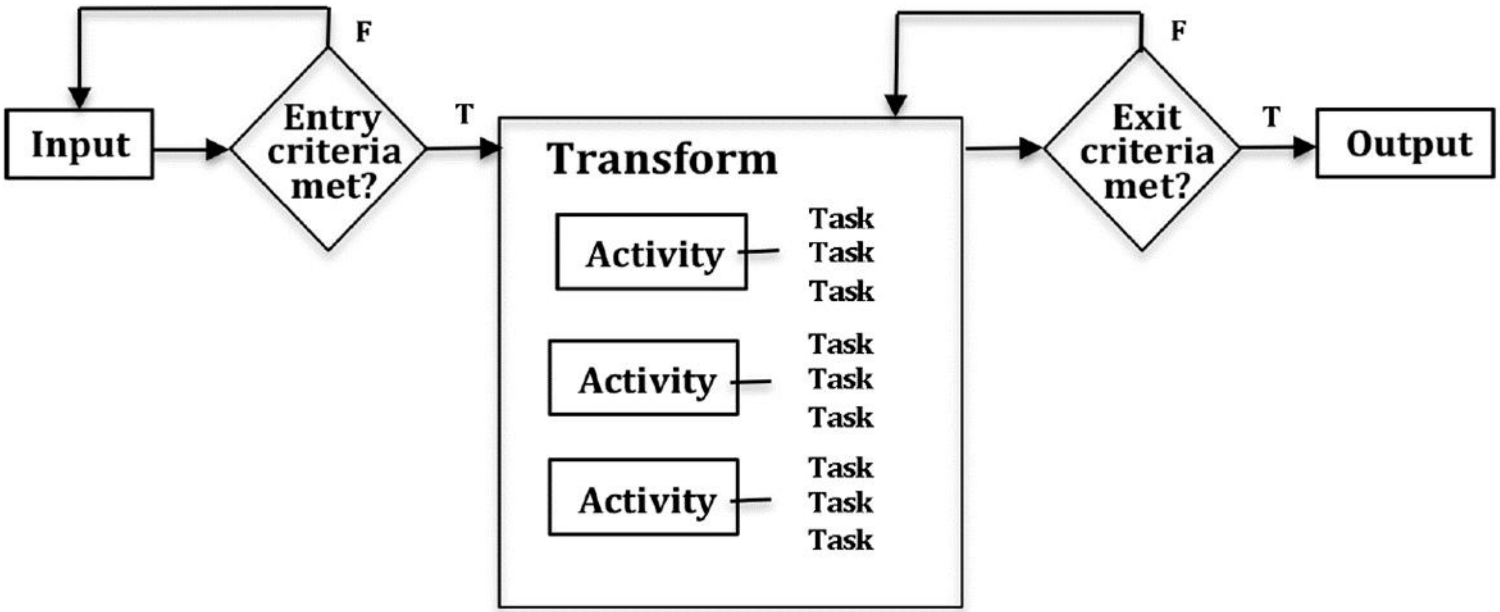
Lesson Ten: Continuous Process Improvement

Outline

- Understand the differences between project management processes and practices
- Understand and be able to construct the Process Quality Matrix (PQM) and the Zone Map
- Have a working knowledge of the Continuous Process Improvement Model (CPIM)
- Know the benefits of having a CPIM
- Be able to diagram a business process
- Know how to use and interpret the eight tools for business process analysis.

Elements of a Software Engineering Process

- **Elements of a Software Engineering Process.**



Elements of a Software Engineering Process

- **Software engineering processes including:**
  - **Primary processes** include software processes for development, operation, and maintenance of software.
  - **Supporting processes** are applied intermittently or continuously throughout a software product life cycle to support primary processes; they include software processes such as software configuration management, software quality assurance, and software verification and validation.
  - **Organizational processes** provide support for software engineering; they include infrastructure management, portfolio and reuse management, organizational process improvement, and management of software life cycle models.
  - **Cross-project processes**, such as reuse and domain engineering; they involve more than a single software project in an organization.

Continuous Process Improvement: Problem vs Solution

- You have invested considerable time in project management process development.
- You have invested considerable money in project management practice training.
- But your projects are still late, over-budget, and not what the client needed. And you can't identify any return on your investment.
- You need to take action, but what action makes sense given what has been done so far?
- **Solution**
- Develop and implement a strategy for the continuous improvement of your project management processes and practices. That strategy called Continuous Process Improvement Model (CPIM).

Continuous Process Improvement: Problem vs Solution

- **What Should the CPIM Give You?**
  - **An improved quality ROI for PM processes.**
  - **Greater alignment between PM processes and their practice.**
  - **A way to leverage best practice.**
  - **A reduced risk of project failure.**
  - **Significant cost avoidance due to increased project success.**

The Project Management Process

- The Process of CPIM
  - How was it developed ?
  - How complete is it ?
  - How is it documented ?
  - How is it supported ?
  - How is it updated ?

The Project Management Practice

- The Practice of CPIM
  - Are all project managers required to use the process?
  - Can project managers substitute other tools, templates, and processes as they deem appropriate?
  - How are project managers monitored for compliance?
  - How are corrective action steps taken to correct for non-compliance?
  - How are project managers practices monitored for best practices?

The Project Management Practice: GQM framework

- **The Goals/Questions/Metrics (GQM) framework can be used to guide process improvement.**
- **An example:**

Goal: to increase the efficiency and effectiveness of software defect discovery and correction during software inspections and reviews.

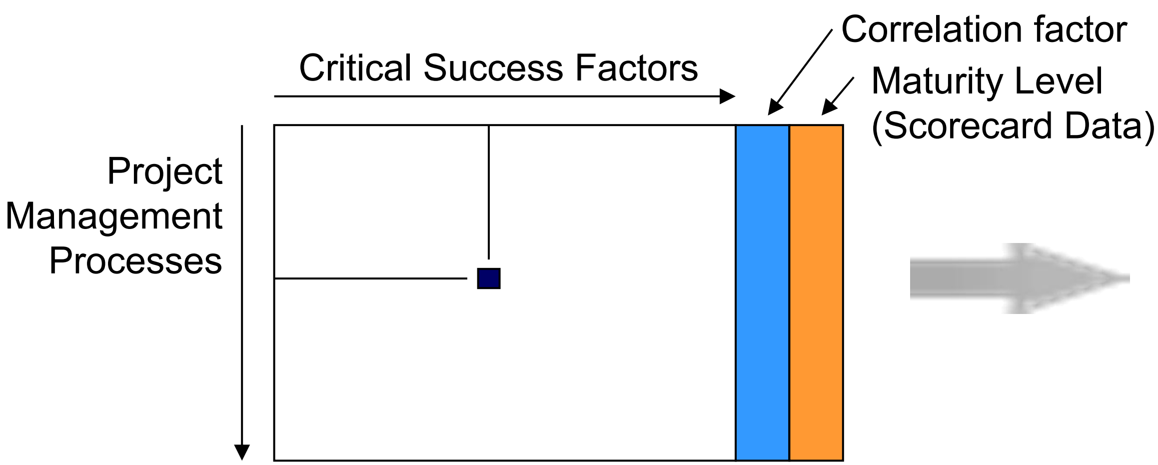
Question: What data is needed to provide insight into enablers and inhibitors of the efficiency and effectiveness of software inspections and reviews.

Metrics:

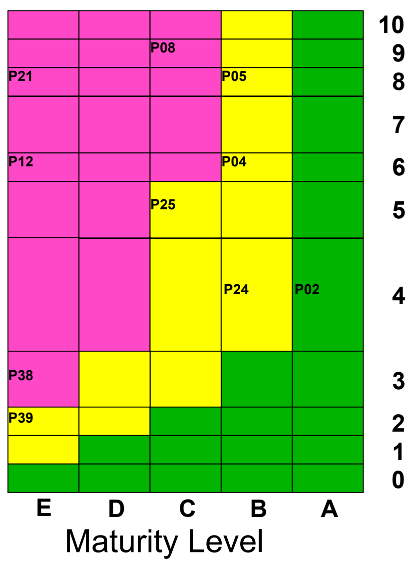
1) frequency of software inspections and reviews;
2) kinds and amounts of material reviewed;
3) skills of reviewers who conduct inspections and reviews;
4) preparation time for reviews and inspections;
5) efficiency and effectiveness of defect discovery and correction.

The Solution – Tools for Continuous Improvement

- **The key inputs to the Continuous Improvement of the Process and Practice of Project Management are the Process Quality Matrix & Zone Map.**

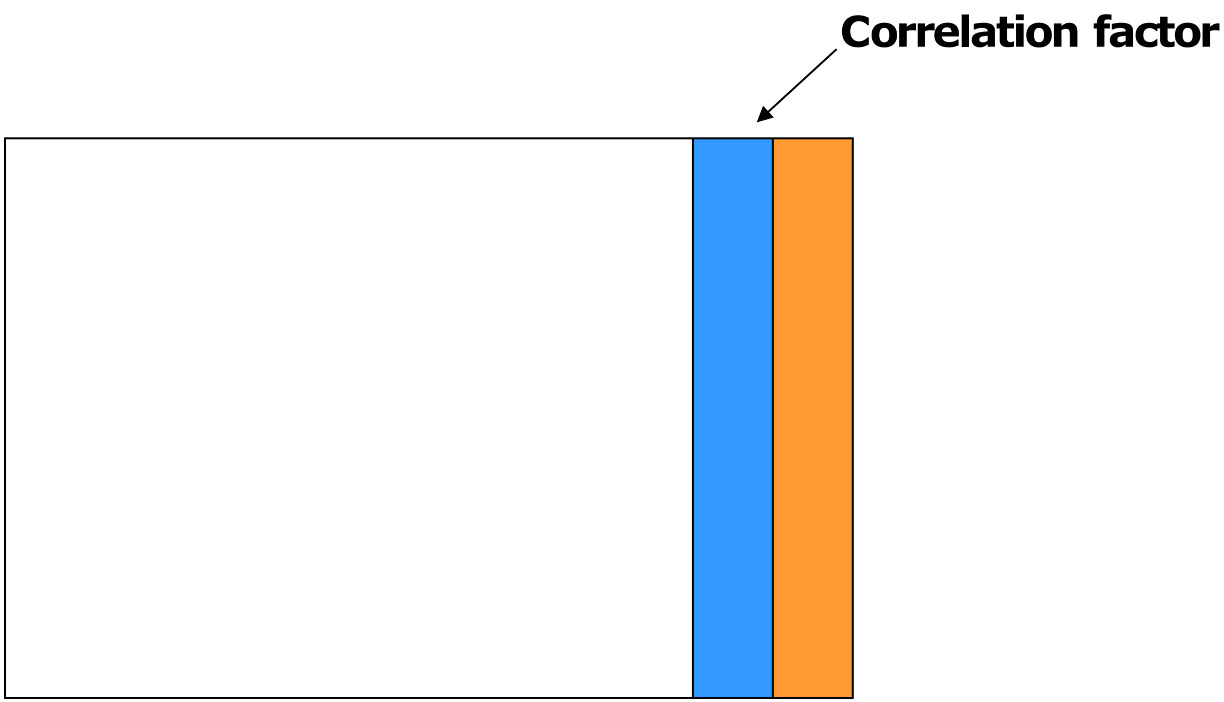


Process Quality Matrix



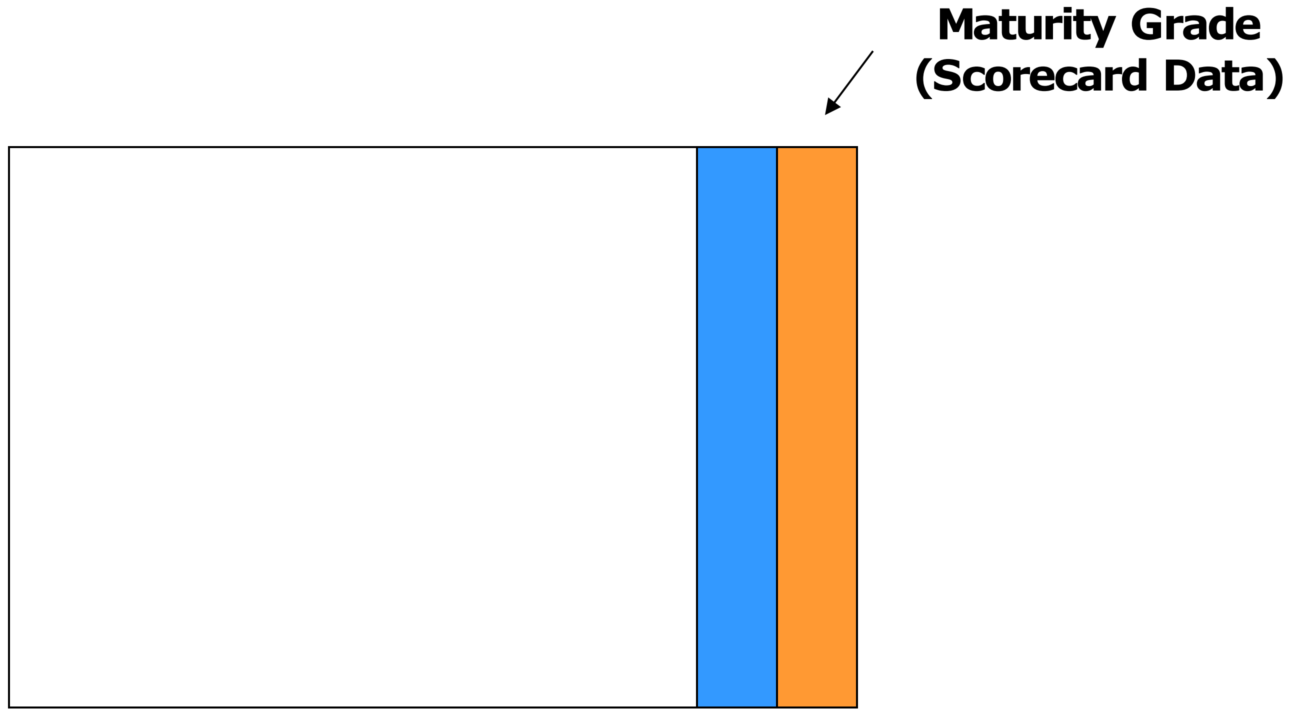
Zone Map

Overview of the Process Quality Matrix



The relationship between the project management processes and the critical success factors

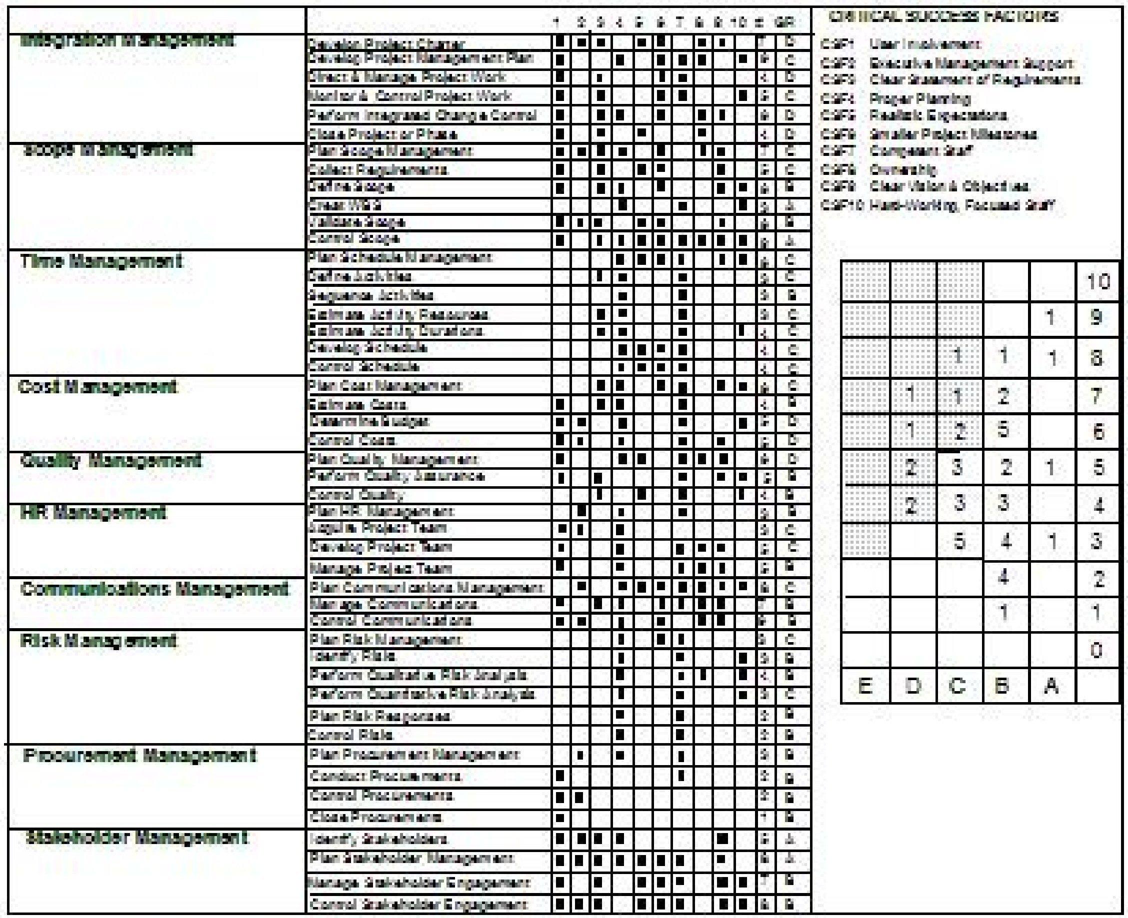
Overview of the Process Quality Matrix



A metric to measure how well a particular process is documented or practiced.

Lesson Ten: Continuous Process Improvement

A completed PQM



Sources: Knowledge Areas: Project Management Institute
Project Management Processes: Project Management Institute
Critical Success Factors: Standish Group CHAOS Report
Process Quality Management: Hardaker & Ward, (1987). HBR Nov-Dec, pp115-118.
Zone Map: Hardaker & Ward

A Completed Zone Map

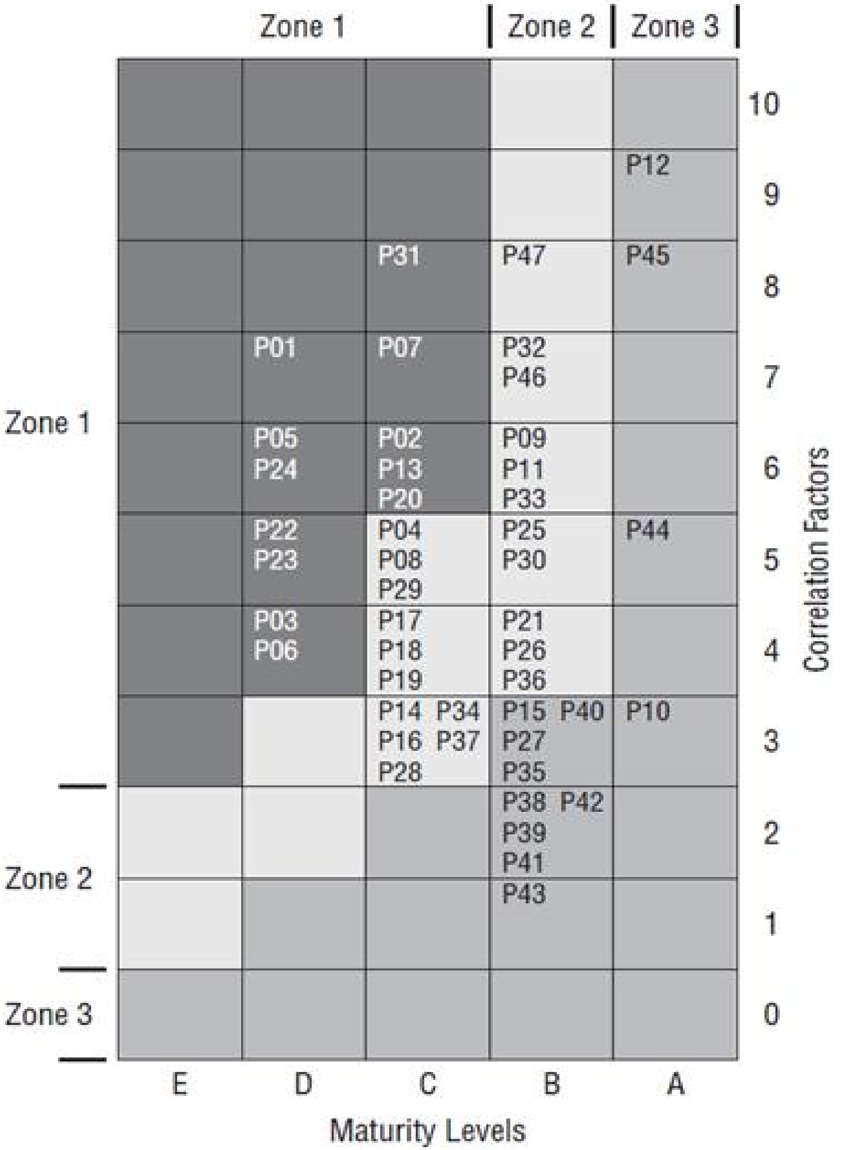


Figure 16-3: A completed Zone Map

Integration Management Zone Map

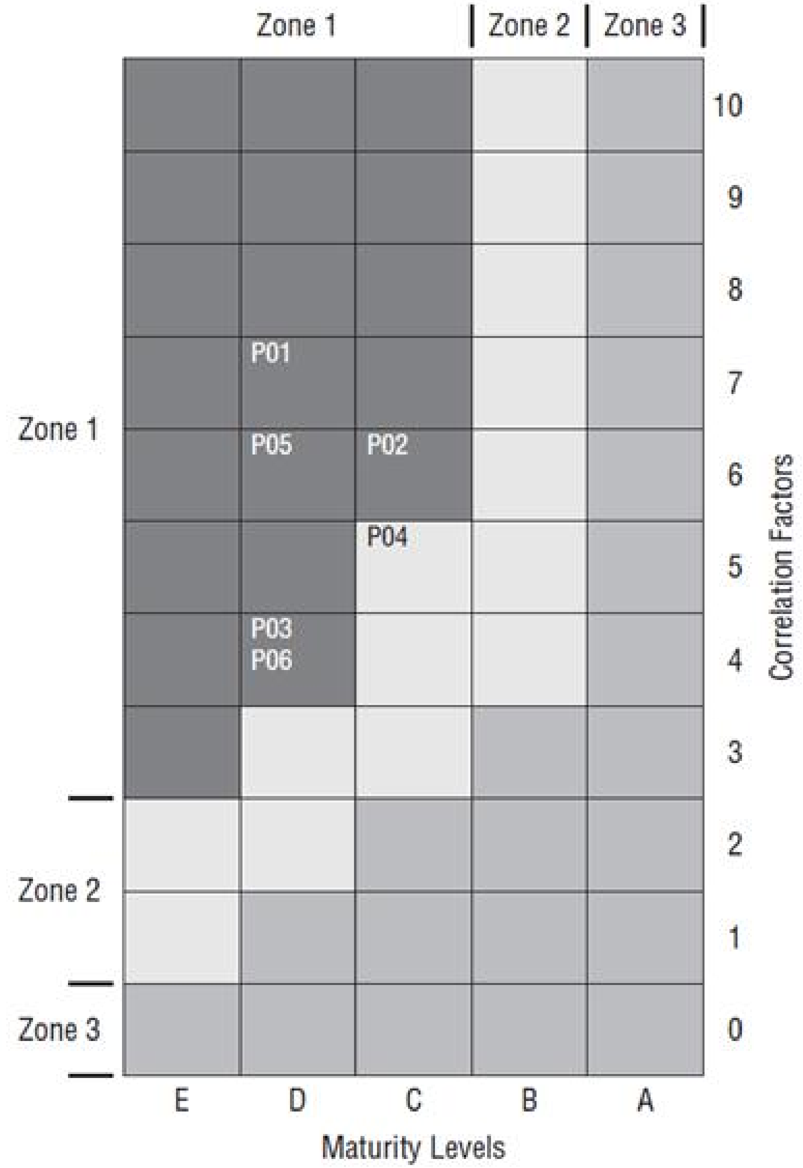


Figure 16-4: Integration Management Zone Map

Planning Management Zone Map

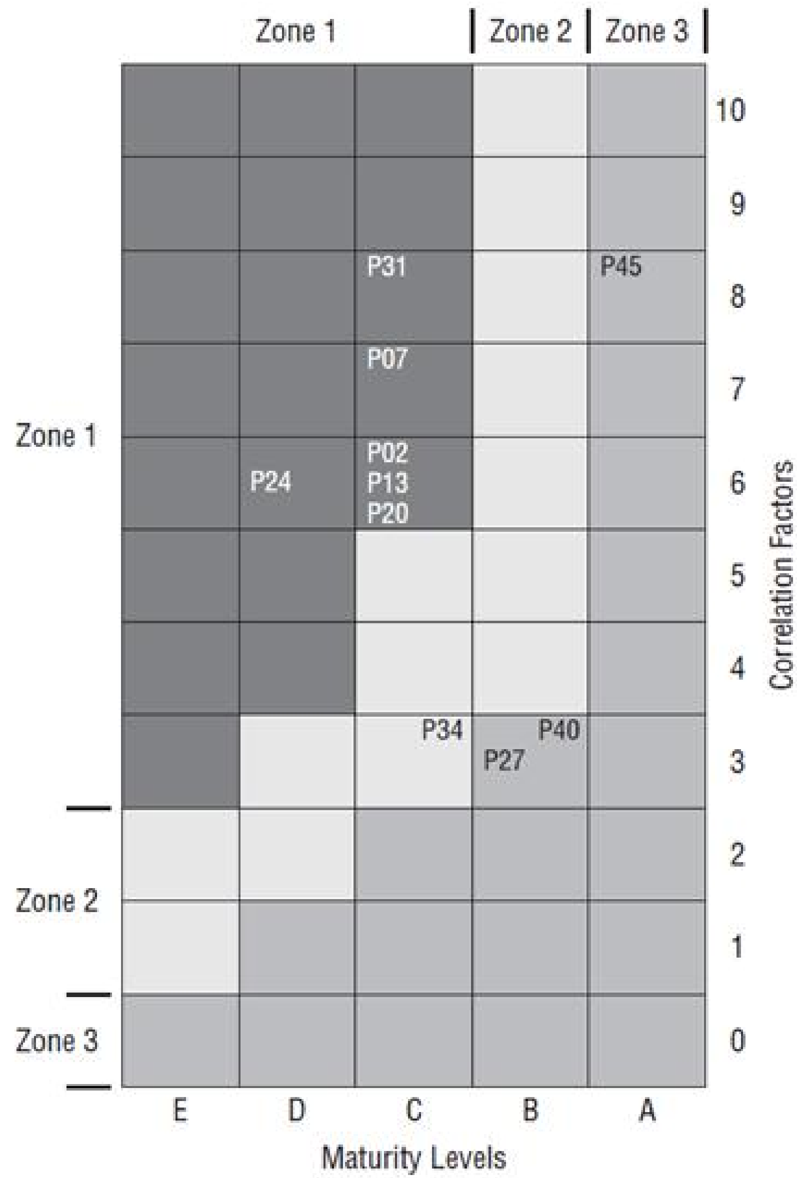
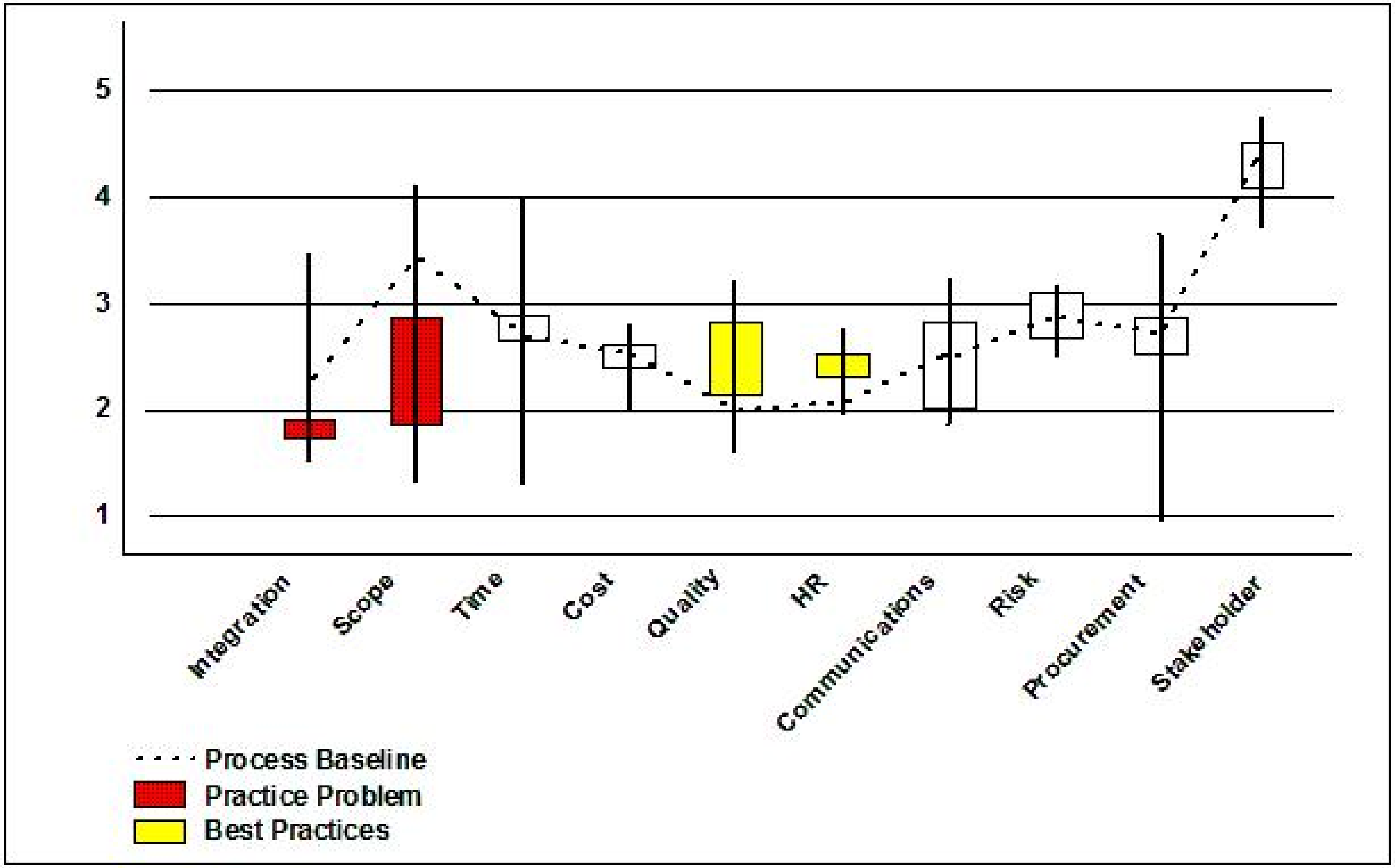


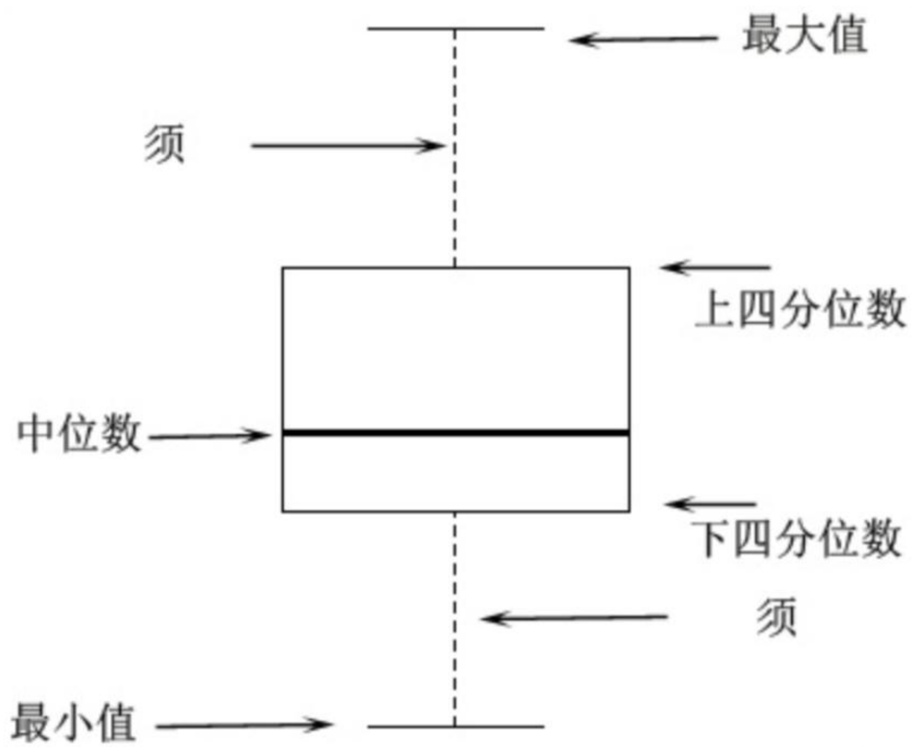
Figure 16-5: Mapping of plan management processes

Process & Practice Maturity Plot



Process & Practice Maturity Plot

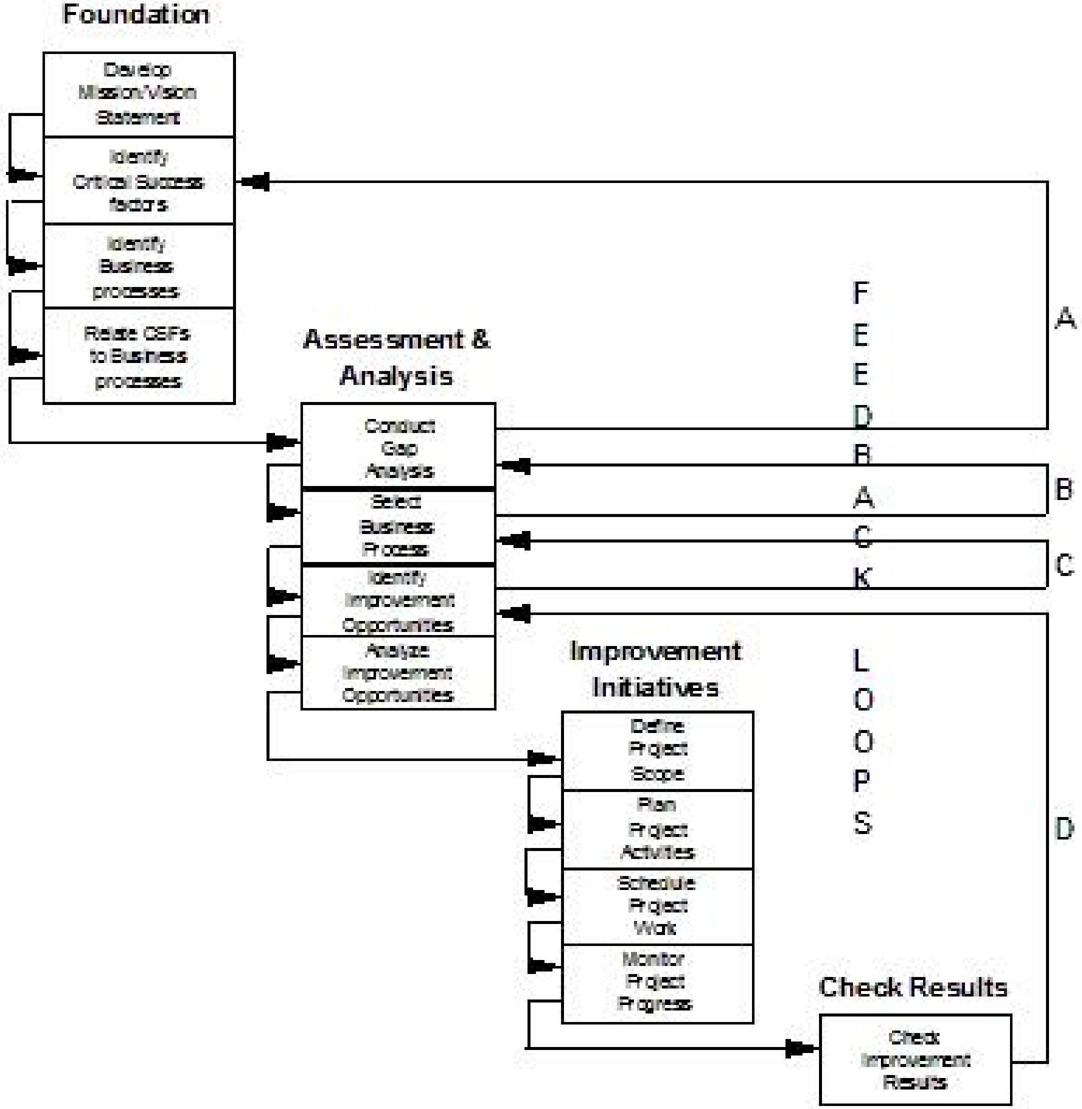
- Box and whisker plot (箱型图)



What Process Has Been Defined So Far?

- Step 1: Define the Process
- Step 2: Validate and finalize PQM
- Step 3: Establish correlations
- Step 4: Establish metrics
- Step 5: Assess project managers against the PMMA
- Step 6: Assess maturity levels
- Step 7: Plot results on the PQM Zone Map

Continuous Process Improvement Model



Using the Continuous Process Improvement Model

- Phase 1: Foundation
  - Develop mission/vision statement
  - Identify Critical Success Factors
  - Identify business processes
  - Relate CSFs to business processes
- Phase 2: Assessment and Analysis
  - Conduct gap analysis
  - Select Knowledge Area of PM Process
  - Identify improvement opportunities
  - Analyze improvement opportunities

Using the Continuous Process Improvement Model

- Phase 2: Assessment and Analysis
  - Analyze improvement opportunities

		VALUE		
		HIGH	MEDIUM	LOW
LEVEL OF EFFORT	High			
	Medium	#3	#4	#2
	Low		#1	

Would you choose initiative #3, and if so, why? Or would you choose initiative #1? Why? Perhaps you could do both #1 and #3 concurrently. Each of these initiatives is independent of the other initiatives, so there won't be a conflict between them.

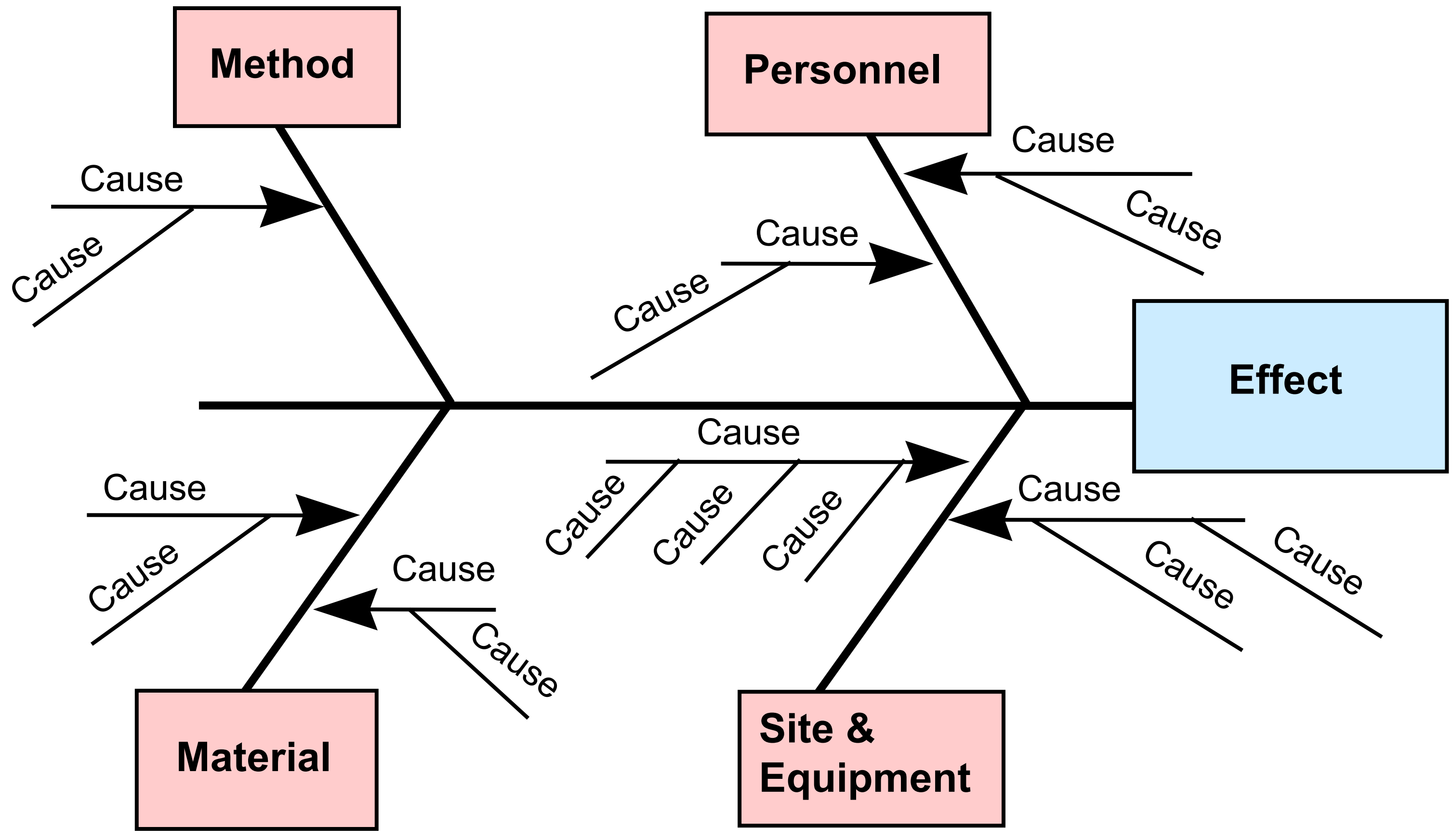
Using the Continuous Process Improvement Model

- Phase 3: Improvement Initiatives
  - Define the project scope
  - Plan project activities
  - Schedule project work
  - Monitor project progress
- Phase 4: Check Results
  - Check improvement results

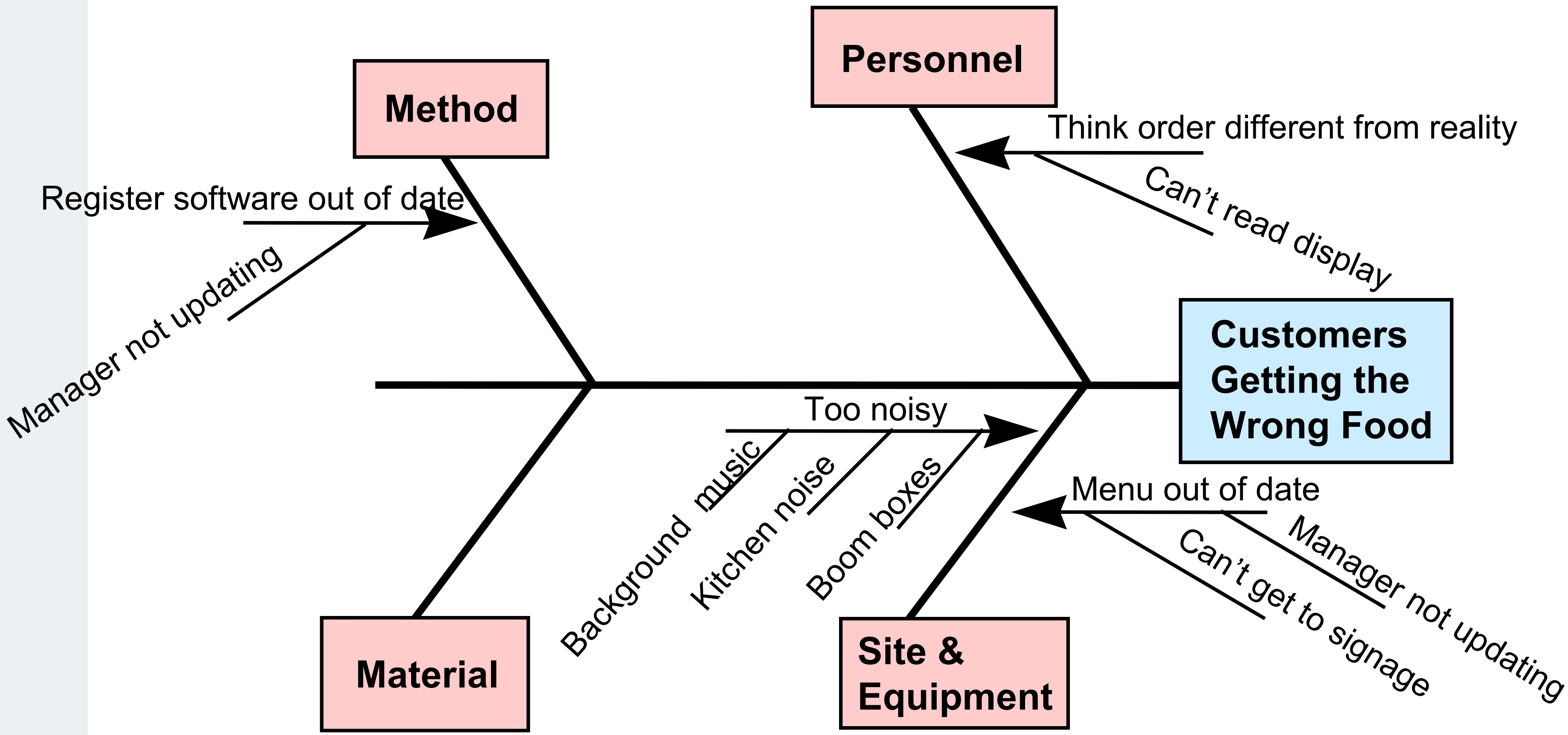
Defining Roles & Responsibilities of the PSO

- Improve the ROI for the PM process
- Have an improvement process in place to significantly reduce the risk of project failure
- Be able to incorporate continuous process improvements
- Enable the PSO to provide leadership in the CPIM
- Have a documented approach that the PSO organization will need to achieve higher PM maturity levels
- Have a process for identifying and proactively acting on the weakest part of the organization's PM capability

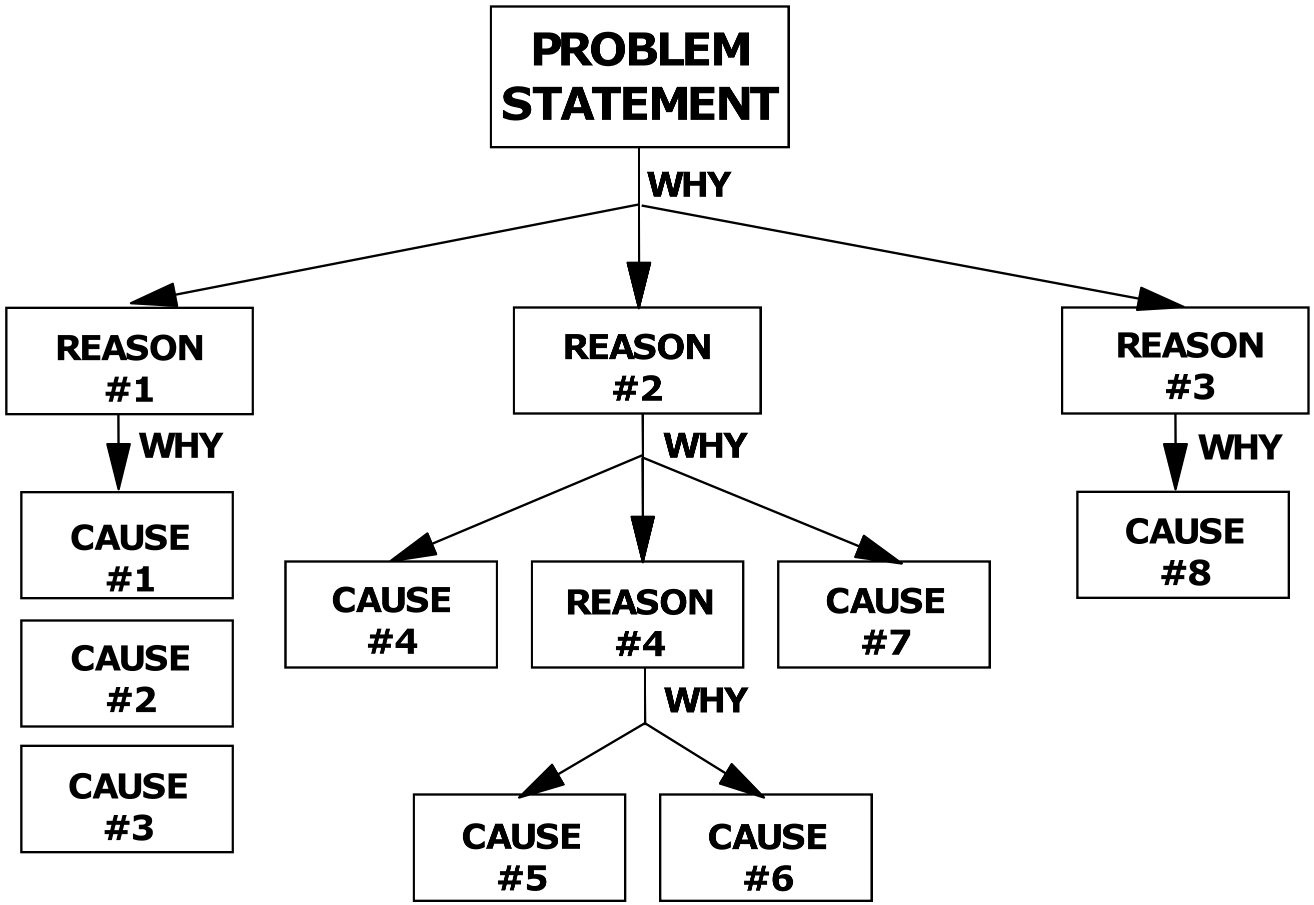
Basic Tool: Fishbone Diagram



Basic Tool: Example – Fishbone Diagram



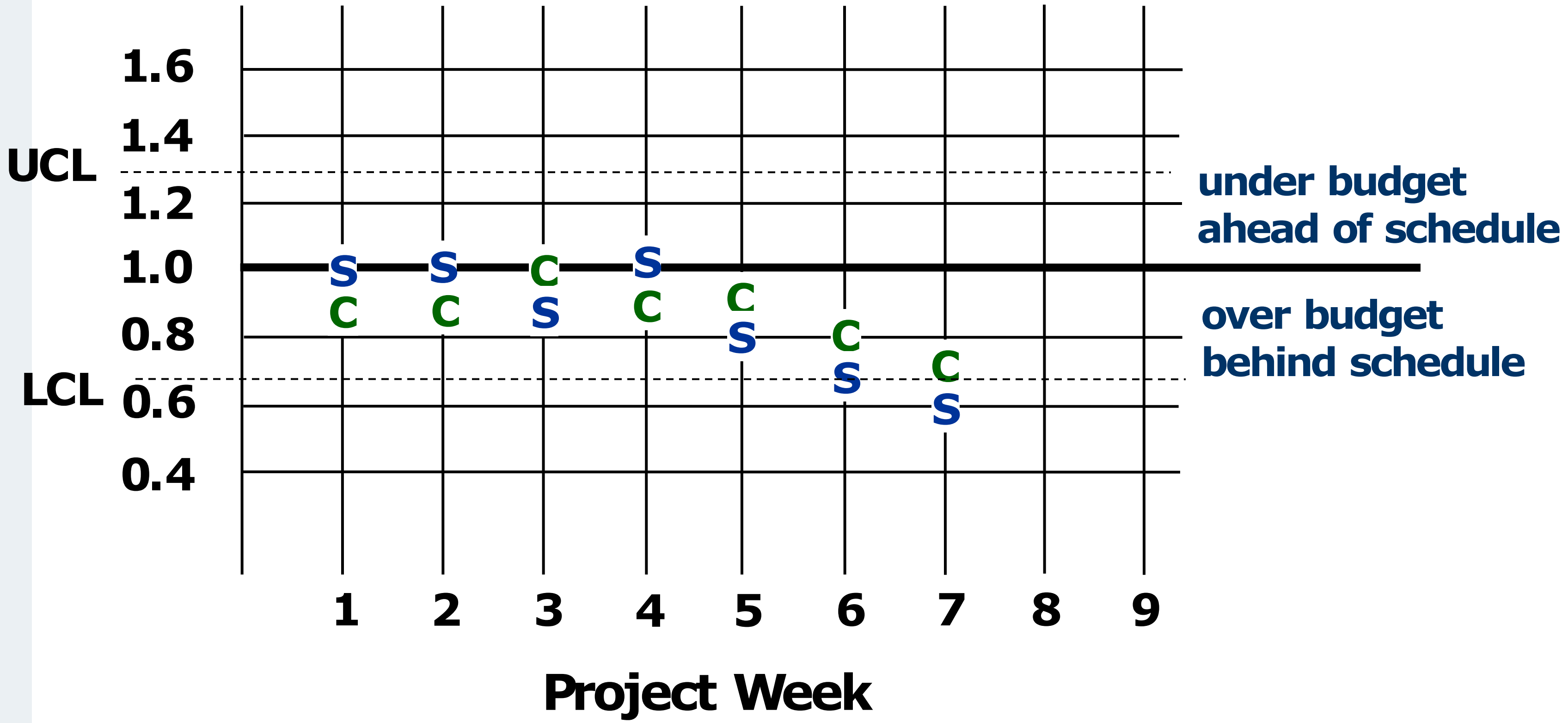
Basic Tool: Fishbone Diagram – Root Cause Template



Basic Tool: Fishbone Diagram – Root Cause Criteria

FISHBONE ROOT CAUSE TEST CRITERIA	YES/NO
Dead End You ran into a dead end when asking, "What caused the proposed root cause?"	
Conversation All conversation has come to a positive end.	
Feels Good Everyone involved is happy, motivated, and emotionally uplifted.	
Agreement All agree it is the root cause that keeps the problem from being resolved.	
Explains The root cause fully explains why the problem exists from all points of view.	
Beginnings The earliest beginnings of the situation have been explored and understood.	
Logical The root cause is logical, makes sense, and dispels all confusion.	
Specific Your statement of the reason gets to the point of the trouble without generalizations.	
Control The root cause is something you can influence, control, and deal with realistically.	
Hope Finding the root cause has returned hope that something constructive can be done about the situation.	
Workable Suddenly, workable solutions that deal with the symptoms begin to appear.	
Stable A stable, long-term, "once and for all" resolution of the situation now appears feasible.	

Basic Tool: Control Charts

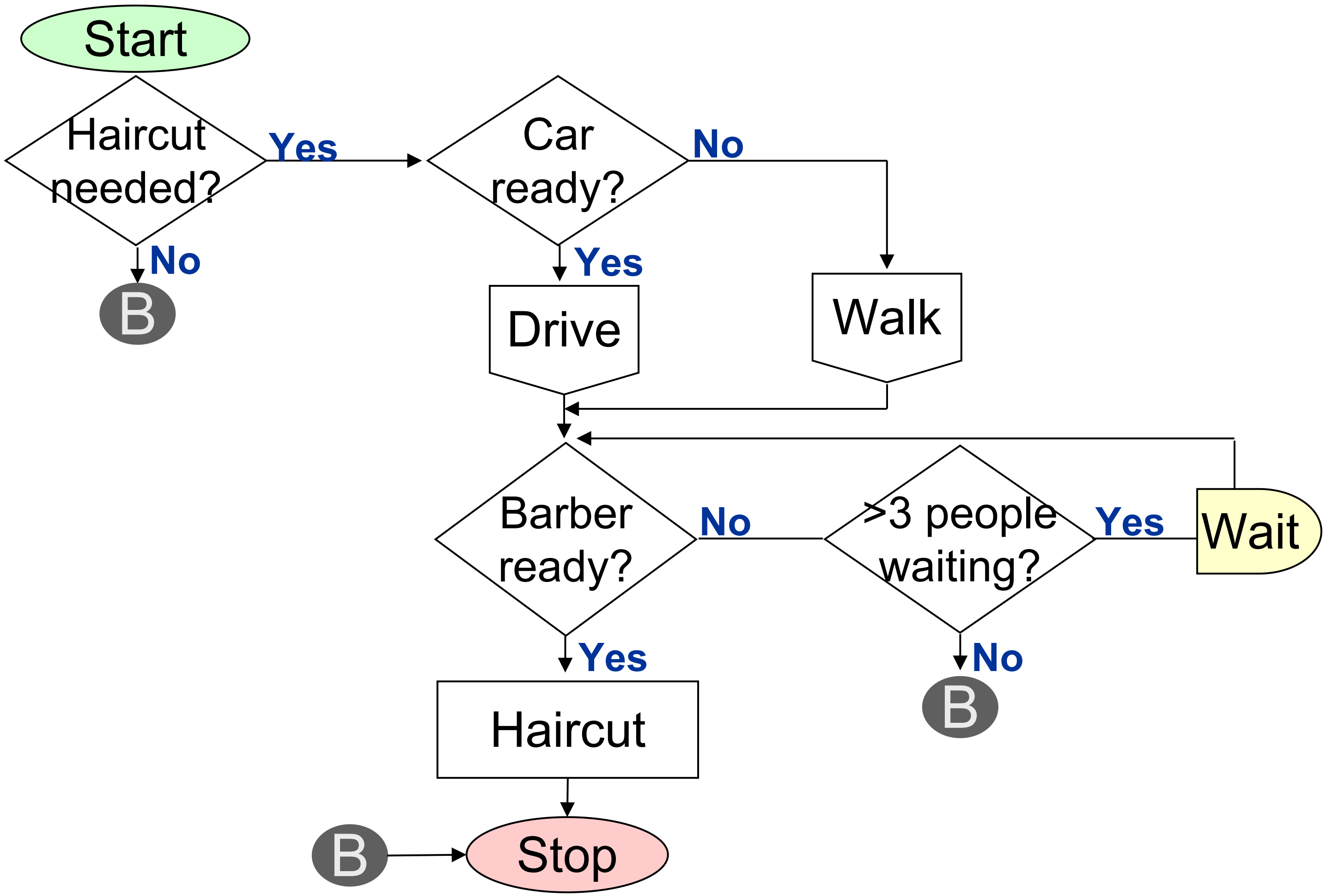


Basic Tool: Flowchart

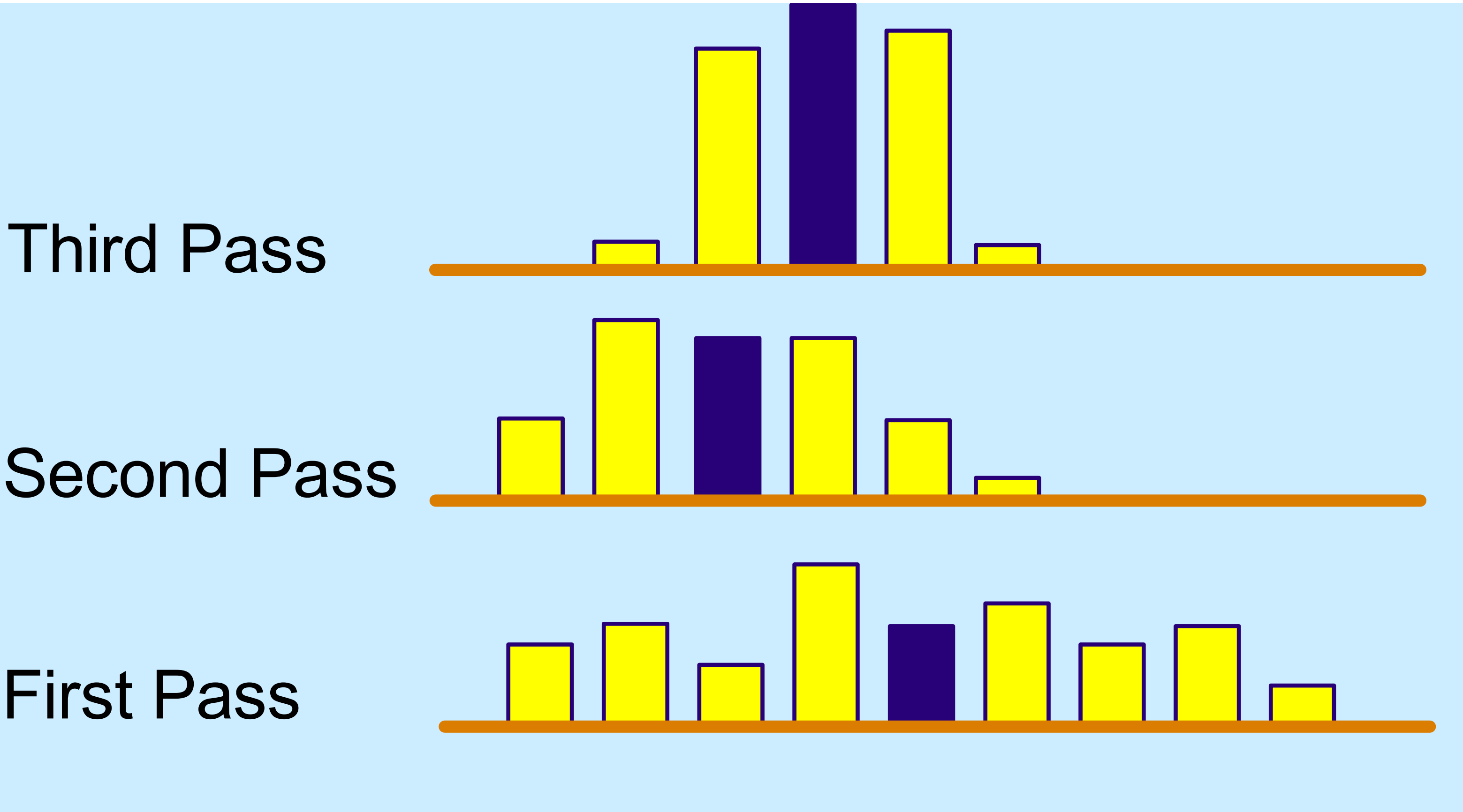
Steps to Create a Flowchart:

- Decide on the process to flowchart
- Define the beginning and ending events of the process
- Describe the beginning event in an oval
- Keep asking "What happens next?" and writing each of the subsequent steps in rectangles below the oval
- When a decision step is reached, write a yes/no question in a diamond and develop each path (make sure that each decision path re-enters the process or is pursued to its logical conclusion)
- Describe the ending event in an oval

Basic Tool: Flowcharting



Basic Tool: Histograms



Basic Tool: Pareto Analysis (帕累托分析)

Steps in the Pareto Analysis:

- Collect data on reasons for the failure or anomaly
- Group the data into similar cause groups
- Count the number of data items in each group
- Plot the histogram from highest to lowest frequency
- Tackle the causes with the highest frequency first

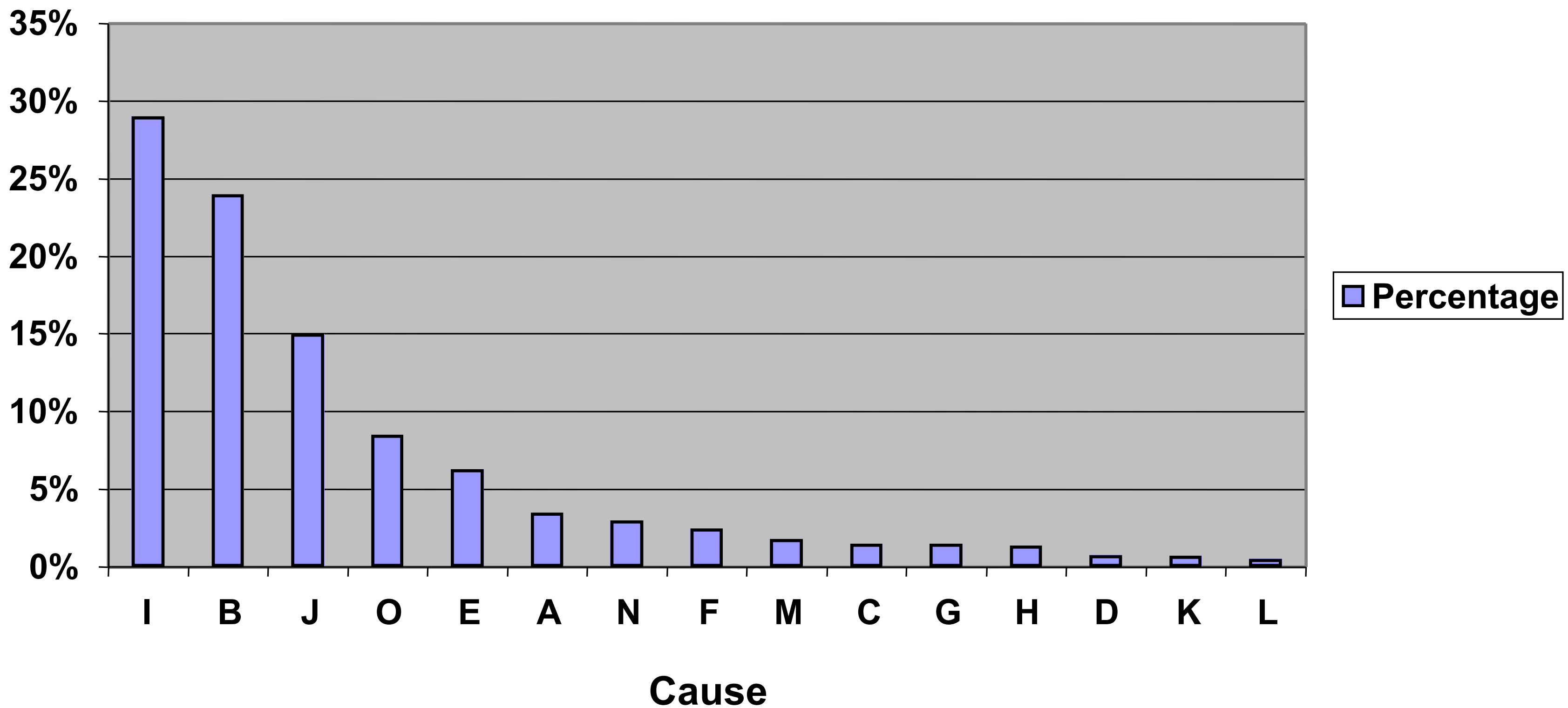
Basic Tool: Pareto Analysis

- We have 15 causes for failure (A – O).
- Which cause do we address first?
  - Pareto = the 80/20 rule
  - Gather the frequencies, sort and chart

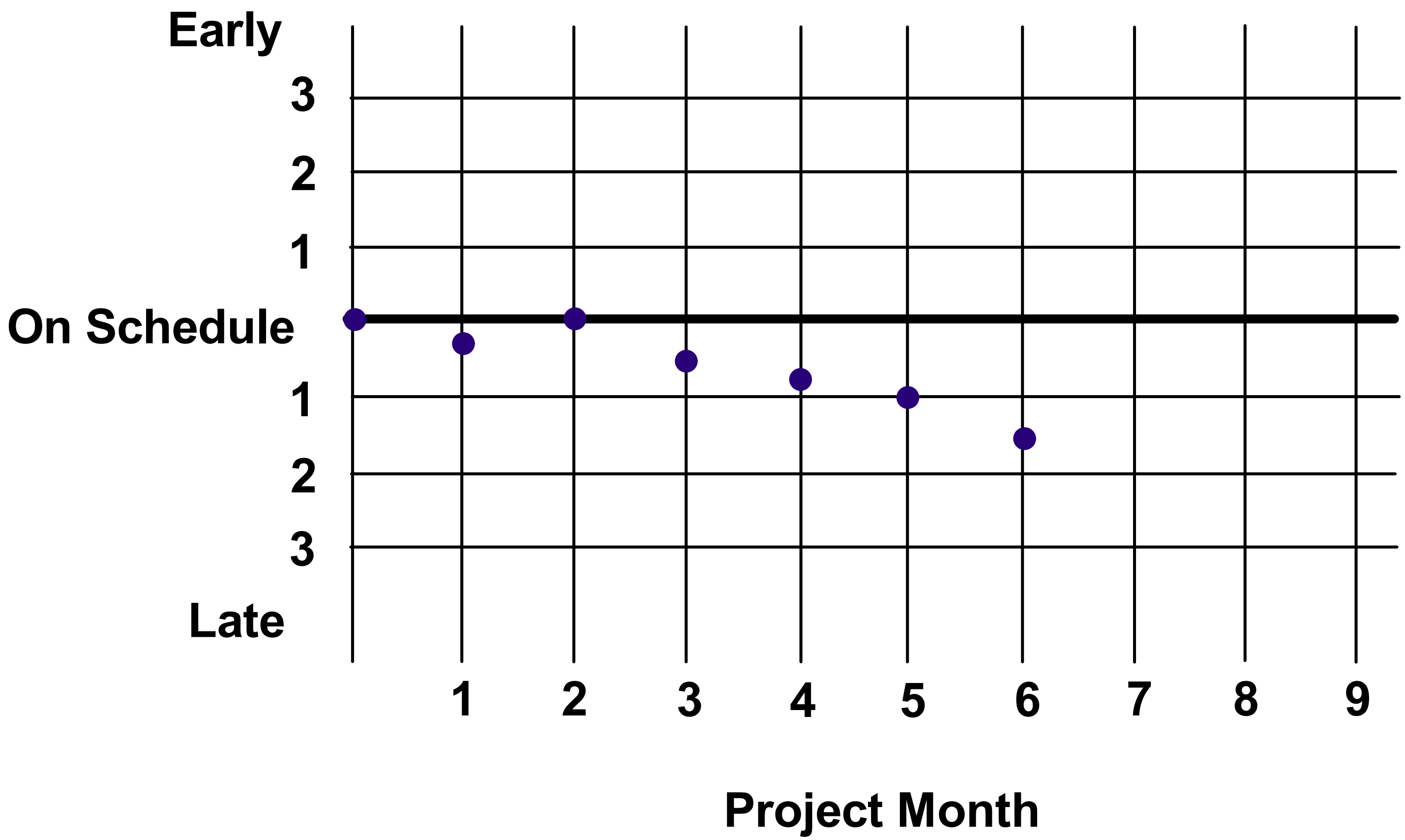
ID	Cause	# Failures
A		35
B	Late arrival	250
C		15
D		8
E		36
F		25
G		15
H		14
I	Missing part	350
J	Access	150
K		7
L		5
M		18
N		30
O		42
		1000

Pareto Analysis

Our top choice is easy to see.

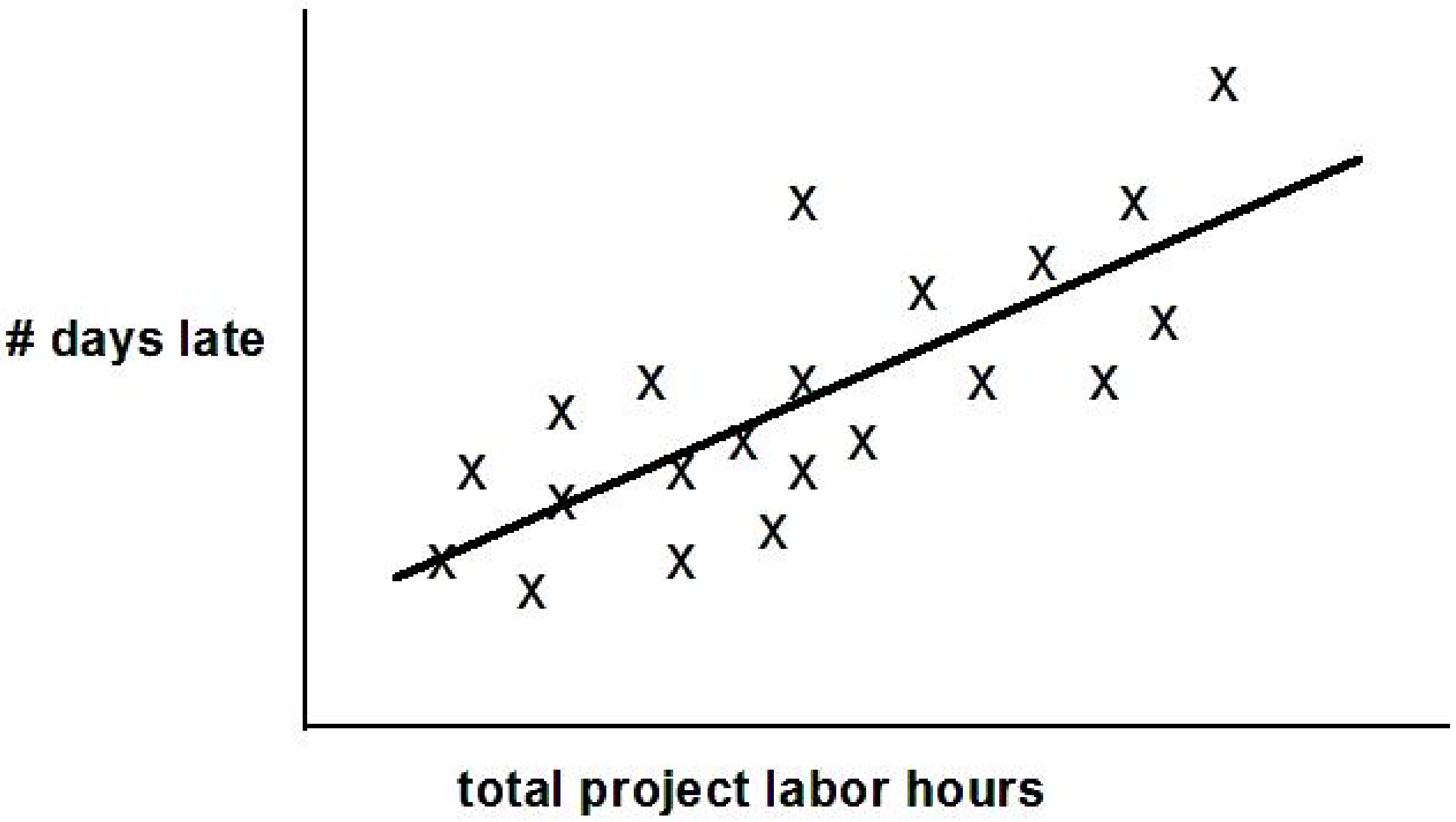


Basic Tool: Run Charts

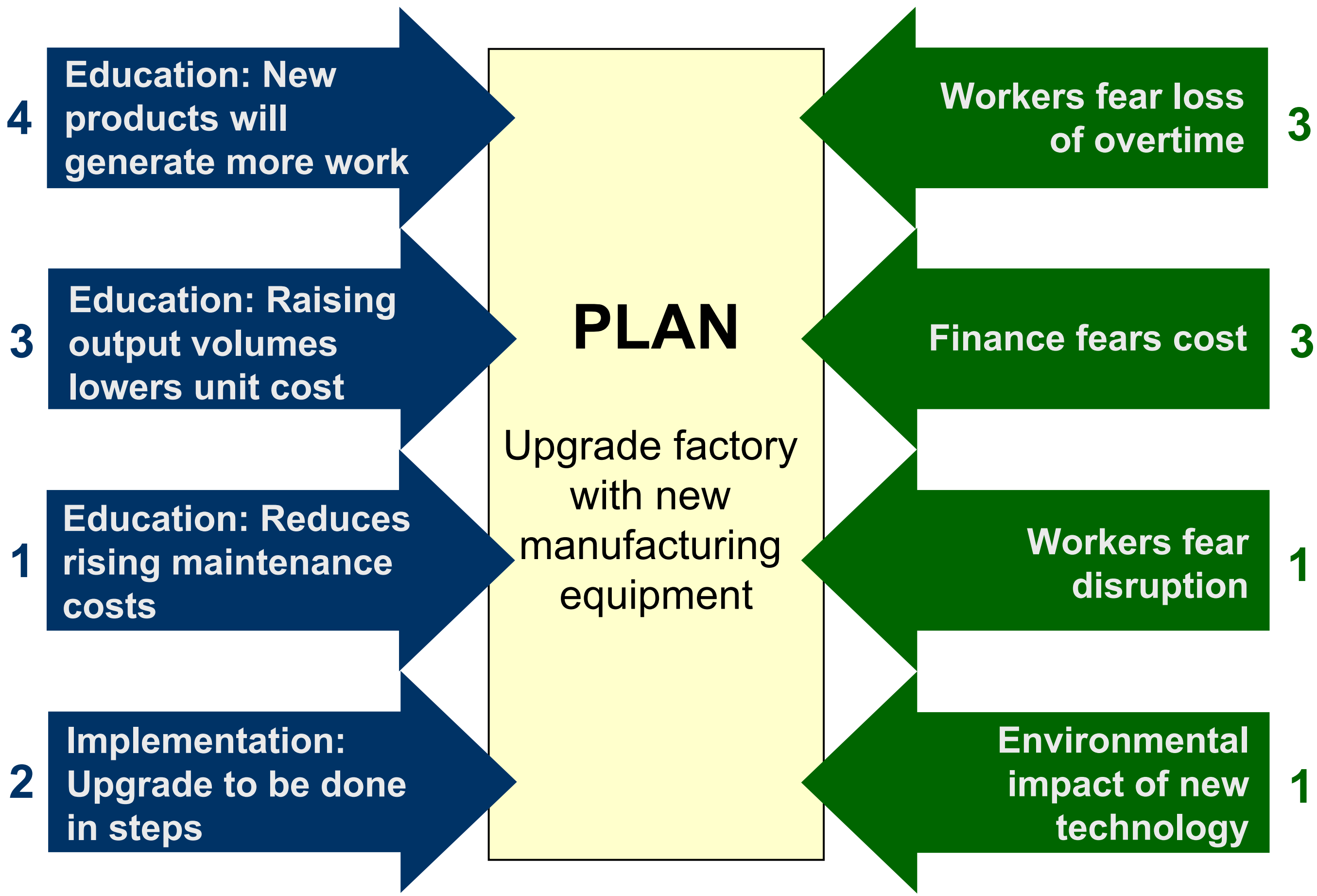


A run up or down of four or more successive data points

Basic Tool: Scatter Diagrams



Bonus Tool: Example – Force Field Analysis



Example - Force Field Analysis – team styles

- Task-oriented team - Management style is the extent to which the execution of the task at hand is paramount.
- People-oriented team - Management style is the extent to which the manager is concerned about the people around them.
- Research shows that where team members are relatively inexperienced, a task-oriented approach is most effective. As group members mature, consideration for their personal needs and aspirations become more valued.
- Where maturity is very high, there is no need for a strong emphasis on either of these approaches.

Summary of Lesson 10

- Lessons learned, project reviews, and other such assessment events are critical input to process and practice improvement. All that is needed is a formal and continuous process for improvement.
- Now you just need the perseverance to make it happen.

Assignment of Lesson Ten

- Submit the **second biweekly report** at **26 April**

- Continually submit the **design document**, the **test document**, the **user manual** and the **source code** to Canva and Github until **14 June**.